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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
Consent: Participants provided written, informed consent for an add-on actigraphy study, described in detail elsewhere.
IRB: This current analysis was approved by the Partners Healthcare, Inc. Institutional Review Board and was part of the Knight ADRC-approved project D1821 (Neuropathology for disrupted multiscale activity control in Alzheimers disease).
Inclusion and Exclusion Criteria
Inclusion criteria were age over 45 years, no cognitive impairment (Clinical Dementia Rating score 0), and no abnormal movement of the nondominant arm.
Attrition
not detected.
Sex as a biological variable
Interpretation: This study showed for the first time a link between FMAR and AD via preclinical pathology in women that was independent of age, race, education, daily activity, and apolipoprotein E e4 status.
Subject Demographics
Age: Inclusion criteria were age over 45 years, no cognitive impairment (Clinical Dementia Rating score 0), and no abnormal movement of the nondominant arm.
Randomization
not detected.
Blinding
not detected.
Power Analysis
not detected.
Replication

Future directions: Future studies may aim to (1) replicate the associations between FMAR and preclinical AD pathology in larger samples of men (2) determine what mechanisms underlie these associations (3) test the relationship between FMAR alterations and other types of dementia and (4) explore whether interventions can modulate FMAR as a way to delay the onset of AD.

Cell Line Authentication

Authentication: Low IV results occur when there is a continuous period of high activity and a continuous period of minimal activity during each day higher IV indicates more fragmentation of the restactivity pattern. 35 2.6 Statistical analysis We examined all continuous variables for normal distribution by visual inspection of histograms and the Kolmogorov-Smirnov test.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Experimental Models: Organisms/Strains			
Biological sex at birth, race (non-Hispanic White, Black, or other), education (years) were self-reported.	non-Hispanic White		
Software and Algorithms			
16 , 21 We included 178 participants (mean age standard deviation (SD) 65.9 8.3 years) who completed the actigraphy study, and had at least one AD biomarker available through the ADRC.	ADRC		
This current analysis was approved by the Partners Healthcare, Inc. Institutional Review Board and was part of the Knight ADRC-approved project D1821 (Neuropathology for disrupted multiscale activity control in Alzheimers disease).	Partners Healthcare		Suggestion: Partners HealthCare Biobank (RRID:SCR_001316)(link)
23 Amyloid beta 42 (Ab42) and phosphorylated tau181 (p-tau) were measured by the ADRC Biomarker Core using enzyme-linked immunosorbent assay (INNOTEST Innogenetics).	ADRC Biomarker		
To ensure good signal quality, recordings were checked using an established MATLAB GUI program (Ver.	MATLAB		Suggestion: MATLAB, RRID:SCR_001622(link)
Statistical analyses were performed using JMP Pro (Ver. 14, SAS Institute).	SAS Institute		Suggestion: Statistical Analysis System, RRID:SCR_008567(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Low IV results occur when there is a continuous period of high activity and a continuous period of minimal activity during each day higher IV indicates more fragmentation of the restactivity pattern. 35 2.6 Statistical analysis We examined all continuous variables for normal distribution by visual inspection of histograms and the Kolmogorov-Smirnov test.	Kolmogorov-Smirnov test
Comparisons between two groups (PiB + and PiB- status) were conducted using independent t tests for normally distributed continuous variables, Mann-Whitney U tests if non-normally distributed, or Chi-squared tests for categorical variables.	t tests
	Mann-Whitney U tests
	Chi-squared tests
Correlations were assessed with Pearsons correlation.	Pearsons correlation

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For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq